ITEC 4366 Advanced Web Development

Midterm Project – API-Driven PHP Website

Project Focus	Build a multi-page PHP website that consumes JSON data from an external API and presents it through a custom Bootstrap-based design.
Primary Skills	PHP template chopping, JSON/API parsing, Bootstrap layout, responsive design, GitHub repository management, AI-assisted development, and video presentation.
Estimated Effort	Medium to High. Start early so you have time to choose an API, test data responses, build pages, polish the interface, and record your walkthrough.
Required Submission	A GitHub repository link and a short video presentation link submitted to the D2L Dropbox.
Required Technology	PHP, Bootstrap 5, HTML/CSS, JSON API data, local development environment, and GitHub.
Point Value	65 points

Background and Purpose

This midterm project connects the skills from Labs 1 through 3 into a more complete web development scenario. Instead of building isolated pages or only reading a local JSON file, you will build a small website that retrieves JSON data from an API, processes that data with PHP, and presents the results through a polished Bootstrap interface.

The professional goal is to practice how modern web developers often work: choosing an available data source, using documentation and AI tools to understand the data structure, building reusable page components, and turning raw data into a usable website for a specific audience.

AI tools are encouraged for this project, but the developer is still responsible for the final result. You may use AI to brainstorm topics, select APIs, explain unfamiliar JSON structures, generate starter PHP code, troubleshoot errors, improve Bootstrap layout, and refine your presentation. However, you must test the code, understand the solution, adapt it to the project requirements, and be able to explain how your website works.

Project Scenario

Choose a topic that interests you and build a small API-driven website around it. Your site should feel like a purposeful mini-application, not just a random data dump. A visitor should understand what your website is about, who it is for, and what useful information the API helps provide.

- A video game discovery site that pulls game/platform data.
- A book finder or reading recommendation site.
- A country information dashboard for travelers, students, or trivia fans.
- A themed catalog, guide, directory, tracker, or resource site built around another approved API.

Suggested APIs

You may choose your own API, but it should be free or reasonably accessible for a student project. Avoid APIs that require payment, complex OAuth workflows, or sensitive personal data. These options are good starting points:

API	Best For	Student Notes	Documentation
TheGamesDB	Video game and platform data	Fun topic for many students, but it may require an API key/account. Good choice if you are comfortable following API key instructions.	https://api.thegamesdb.net/
PokéAPI	Game-like creature, move, item, and ability data	Very beginner-friendly, no authentication required, and AI tools usually understand its structure well.	https://pokeapi.co/docs/v2
Open Library API	Books, authors, ISBNs, covers, and reading data	Good for students who want a book search, a reading list, an author explorer, or a library-style project.	https://openlibrary.org/developers/api
REST Countries	Country facts, flags, languages, currencies, regions, and populations	Simple JSON responses and easy visual design possibilities using cards, tables, filters, and comparison pages.	https://restcountries.com/

Other APIs may be used. When choosing an API, make sure you can successfully retrieve data with PHP before building the rest of the site around it.

Minimum Website Requirements

- **Multi-page PHP website:** Create at least four meaningful pages. Pages may include a home page, search/listing page, detail page, about/project explanation page, comparison page, category page, or another page that fits your idea.
- **Reusable PHP template components:** Use template chopping techniques such as shared header, footer, navigation, configuration, or helper files. Your pages should not repeat the same full HTML structure manually.
- **API integration:** Retrieve JSON data from at least one external API using PHP. You may use file_get_contents(), cURL, or another reasonable PHP approach.
- **Data processing:** Use PHP to decode JSON, loop through records, and display selected fields in a useful way. Your project should show more than one API result when appropriate.
- **User interaction:** Include at least one meaningful interaction such as search, filtering, category browsing, query-string details, sorting, or a user-selected API request.
- **Bootstrap-based design:** Use Bootstrap 5 to create a responsive, professional layout. Custom CSS should be used to apply your own branding, color choices, spacing, and polish.
- **Responsive layout:** The site should work reasonably on desktop and mobile screen sizes.
- **Clear purpose and content:** Include enough written explanation for users to understand the purpose of the site, the data source, and how to use the website.
- **Responsible AI use:** AI may be used heavily, but the submitted project must be tested, adapted, documented, and explainable by you.

GitHub Repository Requirements

- Create a GitHub repository for the project and submit the repository link to D2L.
- The repository should include your PHP, CSS, image/assets, and supporting project files needed to understand your work.
- Add a README.md file that explains the project purpose, the API used, setup instructions, major features, and how AI tools supported your work.
- Do not commit API keys, passwords, database credentials, or other private secrets. If an API key is required, document where the key should be placed using a sample configuration file or placeholder value.
- Your repository history should show meaningful progress rather than one giant final upload if possible.

Short Video Presentation Requirements

Record a short video presentation, approximately 4 to 7 minutes, that demonstrates your website and explains your development choices. The video does not need to be fancy, but it should clearly demonstrate your project's capabilities.

- Briefly introduce your website topic and intended audience.
- Show the GitHub repository and explain the project structure.
- Demonstrate the main pages and features of your website.
- Show where the API data appears in the site and explain how PHP is used to retrieve and display it.
- Explain how reusable template files are used in your project.
- Discuss how AI helped you and what you personally had to test, fix, change, or learn.
- Mention one improvement you would make with more time.

AI Use and Developer Responsibility Clause

AI tools are allowed and encouraged for this project. You may use AI to help with brainstorming, API selection, code generation, debugging, Bootstrap styling, documentation, and presentation planning. However, you are ultimately responsible for the accuracy, functionality, security, originality, and completeness of the work you submit. Submitting AI-generated code that you do not understand, cannot explain, or have not tested is not acceptable. You should be prepared to discuss how your website works, how your API data is retrieved and processed, and what changes you made beyond AI-generated starter material.

Security and Practical Development Notes

- Treat API data as external input. Do not blindly trust it.
- Escape or sanitize output where appropriate before displaying it in HTML.
- Avoid exposing API keys or credentials in GitHub repositories. (ask your AI about this)
- Include basic error handling for failed API requests, empty results, or unexpected data.
- Consider caching small API responses locally if the API has rate limits or performance concerns.
- Keep the project scope realistic. A smaller site that works well is better than a huge idea that barely runs.

Required Submission to D2L

- GitHub repository link.
- Video presentation link or uploaded video file, depending on D2L instructions.

- Optional: a short note if your project requires special setup instructions, an API key, or known limitations.

This project is not about proving you can memorize API syntax. It is about showing that you can use modern development resources, including AI, documentation, examples, and debugging tools, to build a working web application that you understand and can explain.